UNIVERSITÄT BERN

Production of medically relevant radiolanthanides

PhD Thesis submitted by

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Bern, Dean of the Faculty of Medicine

Bern, Dean of the Faculty of Science

Bern, Dean of the Vetsuisse Faculty Bern

To all the people in my life who support me so much.

"Science is magic that works."

Kurt Vonnegut

University of Bern - Faculty of Science - University of Bern

Abstract

Production of medically relevant radiolanthanides

by Edoardo Renaldin

The Thesis Abstract is written here (and usually kept to just this page). The page is kept centered vertically so can expand into the blank space above the title too...

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor…

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ERK **E**xtracellular signal-**R**egulated **K**inases

Introduction

In the recent past decades, the cancer incidence has steadily increased, affecting more than X millions of people. As shown in Figure 1.1A, the incidence rate has been ever on the rise. Nonetheless, the curve of the mortality rate is dropping since the early 1990s, suggesting more effective treatments have been developed and improved over the last 20 years. Nowadays, several methods are employed to heal this disease: surgery, radiotherapy, chemotherapy, hormone therapy, targeted therapy, immunotherapy and cell-based approaches, and nuclear medicine. To achieve an effective removal of this disease, a combination of these techniques is typically applied to treat the patient. However,

Figure XB shows both the incidence and mortality rates per 100 000 cases with respect to the body organ affected.

Tailored treatments is currently the next step to advance further in curing cancer. The cancer relapse still represent the major cause of death.

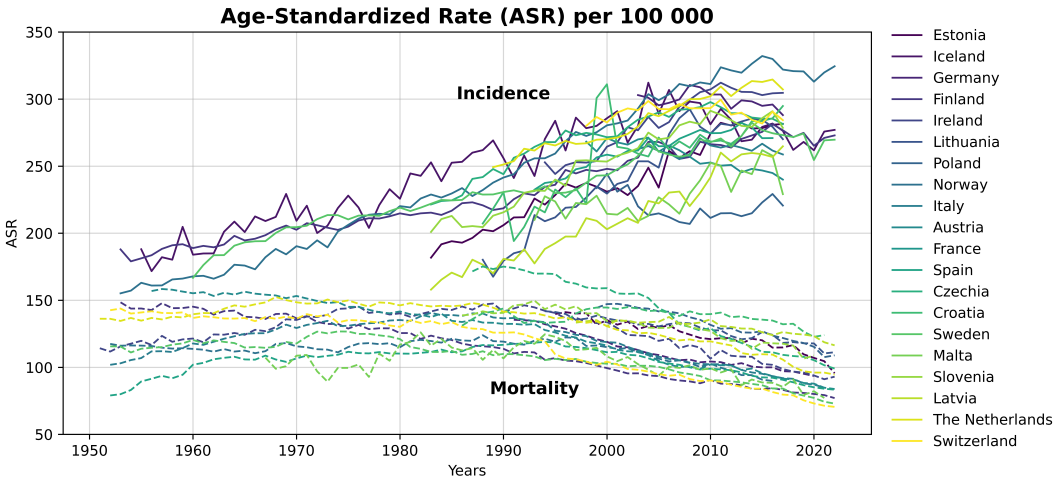

FIGURE 1.1: Cancer age-standardized rate (ASR) of European Countries since the beginning of the second half of the 20th century. Data retrieved from GLOBOCAN 2022 (ref).

1.1 Nuclear Medicine

Intro nuclear medicine

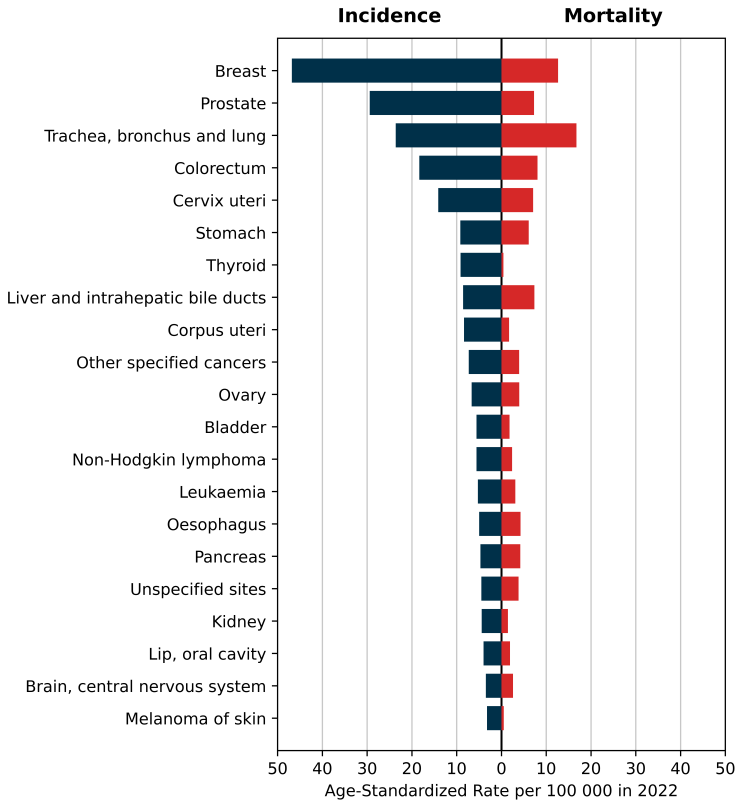

FIGURE 1.2: Incidence and mortality cancer age-standardized rate (ASR) worldwide grouped according to the organs affected. Data retrieved from GLOBOCAN 2022 (ref).

1.1.1 Diagnostic nuclear medicine

1.1.2 Therapeutic nuclear medicine

1.1.3 Theragnostics

1.2 Radiolanthanides in nuclear medicine

1.3 Production of medical radionuclides

1.3.1 Nuclear reactions

1.3.2 Radionuclide production workflow

1.3.3 Cross-section measurements

1.3.4 Liquid column chromatography

Van Deemter equation for calculation of height equivalent to theoretical plate (HETP) relating diffusion, mass transfer, and channeling in packed columns.

Extraction chromatography

Ion exchange chromatography

1.4 Irradiation facilities

The production of medical radionuclides is accomplished through the irradiation of target materials using several projectiles based on the final product. The neutron-rich radionuclides are typically produced using neutron fluxes available in nuclear reactors, while the neutron-deficient ones are obtained using charged particles, such as protons, deuterons, and alpha particles, typically accelerated by cyclotrons. In this work, both protons and neutrons were employed to trigger several nuclear reactions. The different irradiation facilities and their capabilities are described in details in the following subsections.

1.4.1 Proton irradiation

High-intensity proton accelerator (HIPA)

At PSI, a high-intensity cyclotron accelerates protons for multiple research applications. The facility is primarily utilized to generate secondary particles, such as neutrons, pions, and muons, for research in condensed matter physics, materials science, and particle physics **??**. This accelerator was initially named as "Meson Factory" for its initial main application of meson production. However, since its proposal occurred in the 1960s, it underwent continuous upgrades and modifications, becoming one of the most prominent accelerator facility in the world **??**. The overall accelerator is composed of a three-stage accelerator chain:

1. A Cockroft-Walton DC linear accelerator, acting as a pre-accelerator generating a 870 keV proton beam;
2. Injector II cyclotron, an isochronous separated-sector cyclotron increasing the beam energy from 870 keV to 72 MeV in less than 100 turns;
3. Ring cyclotron, the final stage consisting of a massive isochronous machine with eight separate magnets and four main accelerating cavities. This stage accelerated the beam to its final energy of 590 MeV.

IP2 target station

Bern medical cyclotron

1.4.2 Neutron irradiation

Research reactor of the Institut Laue-Langevin

Swiss Neutron Spallation Source (SINQ)

1.5 References

C:/Users/renald$_e$/*Documents/BibTex/library.bib*

List of publications

- **Smith, J.**, Author, A., Author, B., Author, C., and Author, D., 2022. Your hopefully very awesome paper. *Nature* 4.12 (2022).
- Author, A., Author, B., **Smith, J.**, Author, C., and Author, D., 2022. Another very awesome paper you worked on. *Science* 12.53 (2021).

Declaration of Originality

Last name, first name: Renaldin, Edoardo

Matriculation number: 22-136-303

I hereby declare that this thesis represents my original work and that I have used no other sources except as noted by citations.
All data, tables, figures and text citations which have been reproduced from any other source, including the internet, have been explicitly acknowledged as such.
I am aware that in case of non-compliance, the Senate is entitled to withdraw the doctorate degree awarded to me on the basis of the present thesis, in accordance with the "Statut der Universität Bern (Universitätsstatut; UniSt)", Art. 69, of 7 June 2011.

Bern, April 7, 2026

Edoardo Renaldin